

HUJI Applicative Research Report — Week 41, 2026

5 high-potential paper(s) · Weekly · Week 41, 2026 · Generated October 05, 2026

This week's digest highlights groundbreaking research with significant commercial potential across several high-impact fields. Innovations include novel cancer immunotherapies targeting newly discovered micropeptides and advanced liquid biopsy diagnostics for real-time oncology monitoring. Furthermore, we feature a unique, pump-free organ-on-a-chip platform poised to revolutionize drug screening, alongside foundational advances in post-quantum secure distributed computing. Finally, new molecular targets offer promising avenues for precision therapeutics in autism spectrum disorders.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Michal Linial** — Drug Discovery, Vaccines, Genomics, Immunology
- **Ofra Benny** — Drug Discovery, Medical Device, Diagnostics
- **Nir Friedman** — Diagnostics, Genomics, Clinical
- **Ilan Komargodski** — Software/AI, Quantum, Other
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Micropeptides encoded by lncRNAs associated with cancer progression reveal 33/50 novel immunogenic epitopes.

"Novel micropeptides unlock new targets for highly specific cancer vaccines and immunotherapies."

Main researcher: Michal Linial

Department of Biological Chemistry, The Life Science Institute, The Hebrew University of Jerusalem, Jerusalem, Israel.
Published in Bioinformatics (Oxford, England) · 2026-08

ABOUT THIS RESEARCH

Researchers have discovered that long non-coding RNAs (lncRNAs) actually encode small peptides that contribute to cancer progression. These newly identified micropeptides present unique immunogenic epitopes that the immune system can recognize.

COMMERCIAL ANGLE

These micropeptides represent a new class of neoantigen targets for the development of highly specific cancer vaccines and immunotherapies.

COMMERCIALISATION METRICS

Novelty 8/10

The research moves beyond the traditional view of lncRNAs as non-coding regulators to identify functional micropeptides, representing a significant shift in understanding the proteomic landscape for immunotherapy. This discovery of novel immunogenic epitopes provides a high-value, platform-enabling approach for developing new neoantigen vaccines.

Commercial Potential 6/10

The discovery of novel immunogenic epitopes provides a high-value foundation for neoantigen vaccine development or immunotherapy, but the lack of a specific target or delivery platform limits immediate commercial clarity.

Market Size 9/10

The research targets novel immunogenic epitopes derived from lncRNAs, a platform-enabling approach for cancer immunotherapy that addresses the massive, multi-billion dollar oncology market. By uncovering new targets for vaccine and checkpoint development, this work has the potential to expand the addressable market for precision oncology.

Tech Readiness 3/10

The research is at a fundamental discovery stage, identifying novel epitopes through bioinformatics and basic molecular biology without evidence of preclinical animal models or therapeutic formulation. It represents early-stage target identification rather than a developed drug candidate.

IP Strength 7/10

The discovery of novel immunogenic epitopes derived from lncRNA-encoded micropeptides offers strong potential for composition-of-matter patents and platform-based therapeutic development. However, without specific sequence data or a disclosed methodology for targeted delivery, the absolute defensibility against competitive discovery remains moderate.

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2. A pump-free microfluidic device for integrated multi-functional testing of tumor spheroids. 30/50

"Pump-free microfluidic platform offers low-cost, high-throughput drug screening for pharma R&D."

Main researcher: Ofra Benny

The Institute for Drug Research, The School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.
Published in APL bioengineering · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a microfluidic platform that uses passive mechanisms instead of external pumps to study 3D tumor spheroids. This device allows for more integrated and simplified testing of how cancer models react to different environments or drugs.

COMMERCIAL ANGLE

The technology can be commercialized as a low-cost, plug-and-play organ-on-a-chip platform for pharmaceutical companies to improve the accuracy of drug toxicity and efficacy screening.

COMMERCIALISATION METRICS

Novelty

6/10

While the elimination of external pumps simplifies microfluidic architecture, the integration of tumor spheroid testing via passive fluidics is an incremental engineering improvement rather than a paradigm-shifting scientific discovery. The novelty lies in the device's specific configuration for ease of use, but it builds upon well-established principles of capillary-driven or gravity-driven microfluidics.

Commercial Potential

7/10

The device addresses a significant bottleneck in drug discovery (high-throughput 3D modeling) and the 'pump-free' aspect suggests a scalable, cost-effective hardware platform. However, commercial success depends on navigating the highly competitive organ-on-a-chip market and demonstrating clear workflow integration for pharmaceutical R&D.

Market Size

7/10

The technology targets the high-growth organ-on-a-chip and drug discovery market, offering significant value by reducing infrastructure costs and improving high-throughput screening scalability. However, it is a specialized tool within a niche segment of the broader preclinical testing market rather than a broad-scale clinical diagnostic.

Tech Readiness

4/10

The paper describes the fundamental development and proof-of-concept of a microfluidic prototype, which is currently at the laboratory validation stage. While it addresses a significant need in drug testing, it lacks the standardized manufacturing and rigorous biological validation required for near-market or industrial adoption.

IP Strength

6/10

The innovation lies in the physical architecture of a passive microfluidic system, which is patentable, but the lack of a proprietary reagent or biological readout limits its platform defensibility against broader microfluidic competitors.

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3. Plasma nucleosome profiling reports on tumor burden and molecular subtypes28/50 in small cell lung cancer.

"Liquid biopsy platform provides real-time, non-invasive monitoring for lung cancer and personalized treatment."

Main researcher: Nir Friedman

School of Computer Science and Engineering, Hebrew University of Jerusalem, Jerusalem, Israel.; Lautenberg Center for Immunology and Cancer, Institute for Medical Research Israel-Canada, The Hebrew University-Hadassah Medical School, Jerusalem, Israel.; Hebrew University Center for Computational Medicine, The Hebrew University-Hadassah Medical School, Jerusalem, Israel.
Published in The Journal of clinical investigation · 2026-10-01

ABOUT THIS RESEARCH

Researchers have developed a method to analyze nucleosomes in blood plasma to monitor tumor burden and identify molecular subtypes in small cell lung cancer patients. This liquid biopsy approach provides a non-invasive way to track disease progression and characterize cancer genetics through circulating cell-free DNA structures.

COMMERCIAL ANGLE

There is significant commercial potential in developing a high-sensitivity liquid biopsy diagnostic platform for real-time oncology monitoring and personalized treatment selection.

COMMERCIALISATION METRICS

Novelty

5/10

While liquid biopsy applications are high-value, nucleosome profiling is a known methodology in the burgeoning field of cell-free DNA fragmentomics, making this specific application to SCLC an incremental advancement rather than a fundamental breakthrough.

Commercial Potential

8/10

The technology targets high-value clinical unmet needs in oncology, specifically tumor burden and molecular subtyping, which are highly licensable or developable as liquid biopsy diagnostic assays. Profiling nucleosomes in plasma represents a scalable, platform-enabling approach for precision medicine in lung cancer.

Market Size

7/10

Small cell lung cancer represents a high-unmet-need oncology segment, but the specific application to nucleosome profiling for tumor burden may be limited to a niche subset of liquid biopsy monitoring rather than broad-spectrum screening.

Tech Readiness

4/10

The study focuses on profiling nucleosomes in plasma, which represents a liquid biopsy research phase characterizing biomarkers rather than a validated clinical diagnostic tool. While it demonstrates proof-of-concept for tumor burden monitoring, it remains in the analytical validation/discovery stage typical of early-to-mid-stage life science research.

IP Strength

4/10

The innovation appears to be a diagnostic application of existing biological markers (nucleosome profiling) rather than the creation of a novel, patentable molecule or device. Defensibility is likely limited to specific proprietary algorithms or data sets, which are harder to protect than foundational biotechnology.

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4. Scalable distributed agreement from LWE: Byzantine agreement, broadcast, and leader election. 30/50

"Next-gen protocols deliver scalable, post-quantum secure communication for decentralized networks."

Main researcher: Ilan Komargodski

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Givat Ram, Jerusalem, Israel.; NTT Research, Sunnyvale, CA USA.

Published · 2026 Oct 1

ABOUT THIS RESEARCH

This paper proposes a new method for achieving consensus in large networks of computers using Learning With Errors (LWE) cryptography. It allows many different nodes to agree on information efficiently even if some of those nodes are malicious or failing.

COMMERCIAL ANGLE

The research provides a foundation for highly scalable, post-quantum secure communication protocols essential for decentralized data infrastructure and secure multi-party computation.

COMMERCIALISATION METRICS

Novelty

8/10

The paper introduces a significant paradigm shift by replacing standard cryptographic assumptions with Learning With Errors (LWE) to achieve sublinear communication complexity, offering a potentially quantum-resistant path to scalability in distributed consensus. It directly improves upon the state-of-the-art by bridging the gap between computational efficiency and post-quantum security assumptions.

Commercial Potential

7/10

The work addresses a critical bottleneck in scalable blockchain and decentralized infrastructure by leveraging LWE-based cryptography for efficient consensus. While it offers a highly licensable foundational layer for next-generation distributed systems, its commercial value is currently tethered to the broader adoption of post-quantum cryptographic standards in large-scale networks.

Market Size

9/10

Consensus and Byzantine agreement protocols are foundational infrastructure for the entire blockchain and distributed systems market, representing a multi-billion dollar addressable demand. By solving for sublinear communication complexity, this research directly addresses the critical scaling bottleneck that currently limits the global adoption of decentralized financial and enterprise-grade distributed ledgers.

Tech Readiness

3/10

The paper focuses on fundamental cryptographic theory and complexity bounds (LWE-based protocols) rather than a practical implementation or system architecture. It represents early-stage mathematical research that provides the building blocks for future scalable distributed systems but lacks the engineering maturity for immediate commercial application.

IP Strength

3/10

This is a fundamental theoretical contribution in cryptographic protocol design, which typically lacks the specific embodiment or novel physical/methodological utility required for robust, platform-enabling patent protection. While highly impactful for the field of distributed systems, it constitutes mathematical methodology and algorithmic theory that is broadly challenging to ring-fence as proprietary intellectual property.

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5. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling **28/50** by nitric oxide inhibition in the Shank3 mouse model of autism.

"Precision targets identified for novel therapeutic interventions in autism spectrum disorders."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

ABOUT THIS RESEARCH

Researchers used phosphoproteomic profiling to map how nitric oxide inhibition restores normal signaling pathways in a mouse model of autism related to the SHANK3 gene. The study identifies specific protein phosphorylation changes that could explain how targeting nitric oxide synthase can reverse behavioral and synaptic symptoms.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty

6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential

6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size

9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness

3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength

4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

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